

Suppose that we are fitting a Gaussian mixture model for data items consisting of a single real value,  $x$ , using  $K = 2$  components. We have  $N = 5$  training cases, in which the values of  $x$  are as follows:

5, 10, 20, 35, 40

We use the EM algorithm to find the maximum likelihood estimates for the model parameters, which are the mixing proportions for the two components,  $\pi_1$  and  $\pi_2$ , and the means for the two components,  $\mu_1$  and  $\mu_2$ . The standard deviations for the two components are fixed at 10.

Suppose that at some point in the EM algorithm, the E step found that the responsibilities  $\gamma(z_{nk})$  for sample  $n$  and mixture  $k$  are as follows:

|          | Mixture 1 | Mixture 2 |
|----------|-----------|-----------|
| Sample 1 | 0.2       | 0.8       |
| Sample 2 | 0.2       | 0.8       |
| Sample 3 | 0.9       | 0.1       |
| Sample 4 | 0.9       | 0.1       |
| Sample 5 | 0.9       | 0.1       |

(a) What are the values of parameters  $\pi_1$ ,  $\pi_2$ ,  $\mu_1$ , and  $\mu_2$  to be found in the next M step of the algorithm?

$$\mu_1 = \frac{\sum_n \gamma(z_{n1}) x_n}{\sum_n \gamma(z_{n1})} = \frac{0.2 \times 5 + 0.2 \times 10 + 0.9 \times 20 + 0.9 \times 35 + 0.9 \times 40}{0.2 + 0.2 + 0.9 + 0.9 + 0.9} = 28.54$$

$$\mu_2 = \frac{\sum_n \gamma(z_{n2}) x_n}{\sum_n \gamma(z_{n2})} = \frac{0.8 \times 5 + 0.8 \times 10 + 0.1 \times 20 + 0.1 \times 35 + 0.1 \times 40}{0.2 + 0.2 + 0.9 + 0.9 + 0.9} = 11.32$$

$$\pi_1 = \frac{\sum_n \gamma(z_{n1})}{\sum_k \sum_n \gamma(z_{nk})} = \frac{0.2 + 0.2 + 0.9 + 0.9 + 0.9}{5} = 0.62$$

$$\pi_2 = \frac{\sum_n \gamma(z_{n2})}{\sum_k \sum_n \gamma(z_{nk})} = \frac{0.8 + 0.8 + 0.1 + 0.1 + 0.1}{5} = 0.38$$

(b) What are the new responsibilities  $\gamma(z_{nk})$ ?

The new responsibilities for each sample belonging to Mixture 1 are:

$$\gamma(z_{11}) \propto \pi_1 \mathcal{N}(x_1 | \mu_1, 10) = 0.62 \times \mathcal{N}(5 | 28.54, 10) = 6.973 \times 10^{-14}$$

$$\gamma(z_{21}) \propto \pi_1 \mathcal{N}(x_2 | \mu_1, 10) = 0.62 \times \mathcal{N}(10 | 28.54, 10) = 2.623 \times 10^{-14}$$

$$\gamma(z_{31}) \propto \pi_1 \mathcal{N}(x_3 | \mu_1, 10) = 0.62 \times \mathcal{N}(20 | 28.54, 10) = 0.00203$$

$$\gamma(z_{41}) \propto \pi_1 \mathcal{N}(x_4 | \mu_1, 10) = 0.62 \times \mathcal{N}(35 | 28.54, 10) = 0.00968$$

$$\gamma(z_{51}) \propto \pi_1 \mathcal{N}(x_5 | \mu_1, 10) = 0.62 \times \mathcal{N}(40 | 28.54, 10) = 0.000109$$

The new responsibilities for each sample belonging to Mixture 2 are:

$$\gamma(z_{12}) \propto \pi_2 \mathcal{N}(x_1|\mu_2, 10) = 0.38 \times \mathcal{N}(5|11.32, 10) = 0.0162$$

$$\gamma(z_{22}) \propto \pi_2 \mathcal{N}(x_2|\mu_2, 10) = 0.38 \times \mathcal{N}(10|11.32, 10) = 0.0004501$$

$$\gamma(z_{32}) \propto \pi_2 \mathcal{N}(x_3|\mu_2, 10) = 0.38 \times \mathcal{N}(20|11.32, 10) = 1.9012 \times 10^{-10}$$

$$\gamma(z_{42}) \propto \pi_2 \mathcal{N}(x_4|\mu_2, 10) = 0.38 \times \mathcal{N}(35|11.32, 10) = 3.6542 \times 10^{-28}$$

$$\gamma(z_{52}) \propto \pi_2 \mathcal{N}(x_5|\mu_2, 10) = 0.38 \times \mathcal{N}(40|11.32, 10) = 3.0394 \times 10^{-36}$$